LESSON 10.3

Math is Playful

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can use patterns and relationships to make sense of the Tower of Hanoi puzzle and think logically about its solutions.

WHY THIS MATTERS The Tower of Hanoi is a puzzle invented in 1883. It has three rods and a stack of discs, smallest on top to largest on the bottom. The number of discs can vary, but the number of rods does not — and hiding inside it is one of the prettiest patterns in mathematics.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
puzzle *Spanish: rompecabezas*	A problem designed to be played with — it has rules, a goal, and room to imagine solutions.	Tower of Hanoi: three rods, a stack of discs, and two rules.
pattern *Spanish: patrón*	Something that repeats or changes in a predictable way.	Circle, square, triangle, circle, square, triangle …
pattern rule *Spanish: regla del patrón*	A rule that tells you how to get the next value in a pattern from the one before it.	1 → 3 → 7 → 15: the rule is “double it and add 1”.
predict *Spanish: predecir*	To say what will happen before it happens, using a pattern or reasoning.	We counted towers of 1, 2 and 3 discs. Reach PAST them: the rule says 5 discs takes 31 steps — before moving one.
organize *Spanish: organizar*	To arrange information — often in a table — so patterns become easier to see.	Discs 1, 2, 3 in one column and steps 1, 3, 7 in the next.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Before moving anything, I describe what the solved puzzle will look like: the discs in the same order — smallest on top to largest on bottom — but on a different rod.

1. Then I play the small cases: a one-disc tower takes 1 move, and a two-disc tower takes 3 moves.
2. A three-disc tower takes at least 7 moves. Small cases are how mathematicians play.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Discs: 1, 2, 3 — Steps: 1, 3, 7. What is happening from each row to the next?

1. _____
2. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The rule is _____, so _____ is legal / breaks it because _____.
- The number of _____ never changes.
- There are always _____ .
- Being playful with math means _____ .

Word bank puzzle | pattern | pattern rule | predict | organize

WATCH OUT!

A common mistake

Doubling the previous number of steps but forgetting to add 1

LESSON 10.3

Math is Playful

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1. Then I play the small cases: a one-disc tower takes 1 move, and a two-disc tower takes 3 moves.
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1 **Circle the letter of the best answer. Show how you know.**

What is the fewest number of moves needed to solve a three-disc Tower of Hanoi puzzle?

(A) 3 moves
(B) 6 moves
(C) 7 moves
(D) 9 moves

STEPS TO TRY
1. It is more than one move per disc.
2. A two-disc tower already takes 3 moves.
3. Double 3 and add 1.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Which move is against the rules of the Tower of Hanoi?

(A) Moving one disc at a time

(B) Moving the top disc from a stack

(C) Placing a larger disc on top of a smaller disc

(D) Placing a smaller disc on top of a larger disc

STEPS TO TRY

1. Two of these are the rules themselves.
2. Think about disc SIZES.
3. Which stacking order would crush a smaller disc?

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

What will the solved puzzle look like?

(A) The discs in the same order — smallest on top to largest on bottom — but on a different rod

(B) The discs in reverse order, largest on top

(C) The discs spread out with one disc on each rod

(D) The discs back on the starting rod in any order

STEPS TO TRY

1. Imagine the ending before you start.
2. The stack looks just like the start.
3. Only its rod has changed.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

How many moves does a one-disc tower take? A two-disc tower?

(A) 1 move and 3 moves

(B) 1 move and 2 moves

(C) 1 move and 4 moves

(D) 2 moves and 4 moves

STEPS TO TRY

1. Play the one-disc game in your head first.
2. For two discs: the small one has to get out of the way, and later get back on top.
3. Count those three separate moves.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

Take a move you called legal and your partner called a rule-break. What is the rule in your own words, and does it still hold on a move you both agree about?

SENTENCE STARTER The rule is ___, so ___ is legal / breaks it because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help